
VERIFIED PEPTIDES NETWORK

Complete Peptide Reference Database

Dosing · Mechanisms · Duration · Protocols · Research Studies · Stacks

79 COMPOUNDS · PINNED LANDMARK STUDIES · INTEL FEED HIGHLIGHTS · BLOOD WORK REFERENCE

FOR RESEARCH & EDUCATIONAL PURPOSES ONLY · NOT MEDICAL ADVICE · NOT FOR HUMAN CONSUMPTION

All compounds listed are research chemicals only. Nothing in this document constitutes medical advice, treatment recommendations, or endorsement of human use. Always consult a licensed physician. Do not begin any protocol without proper medical supervision.

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1. Safety & Disclaimer

■ All content is for research and educational purposes only.

All compounds referenced in this document are research chemicals only. They are sold and used exclusively for in-vitro laboratory research, preclinical animal studies, and scientific investigation. They are NOT intended, approved, or marketed for human therapeutic use, human consumption, or veterinary use unless explicitly FDA-approved.

GLP-1 Compound Risks — Must Read Before Any Research

Medullary Thyroid Carcinoma (MTC) / MEN 2	People with personal or family history of MTC or Multiple Endocrine Neoplasia Type 2 MUST NOT use GLP-1 agents — FDA boxed warning. Absolute contraindication.
Diabetic Retinopathy	Rapid blood sugar reduction from GLP-1 agents has been linked to worsening of diabetic retinopathy. Diabetics with pre-existing retinopathy face serious vision loss risk.
Pancreatitis History	History of pancreatitis is a significant contraindication. GLP-1 agents are associated with pancreatic inflammation and life-threatening recurrence risk.
Purity & Sourcing	Sourcing peptides outside licensed compounding pharmacies carries serious risks: unknown purity, incorrect dosing, contamination, no sterility guarantee. Always verify COAs from independent labs (e.g., Janoshik) before any research use.

Baseline blood work is non-negotiable before any research protocol. Thyroid, inflammation markers, hormones, and a full liver panel are minimum baselines.

2. GLP-1 / METABOLIC COMPOUNDS

Retatrutide

Triple agonist (GLP-1/GIP/Glucagon) — next-gen weight loss

DOSE	4–12 mg SC weekly
DURATION	24–48 weeks typical protocol
PROTOCOL	Start 2 mg, titrate up every 4 weeks; take same day each week

Tirzepatide

Dual GLP-1/GIP agonist — powerful weight loss and metabolic health

DOSE	2.5–15 mg SC weekly
DURATION	72 weeks (SURMOUNT-1 trial)
PROTOCOL	Start 2.5 mg, increase by 2.5 mg every 4 weeks to target dose

Semaglutide

GLP-1 agonist — weight loss and blood sugar control

DOSE	0.25–2.4 mg SC weekly
DURATION	68 weeks+ (STEP trials)
PROTOCOL	Ramp 0.25 mg → 0.5 → 1 → 1.7 → 2.4 mg, each step 4 weeks

Cagrilintide

Amylin analog — appetite suppression and weight management

DOSE	0.16–2.4 mg SC weekly
DURATION	26+ weeks (often combined with semaglutide)
PROTOCOL	Currently studied in combination with semaglutide (CagriSema)

Mazdutide

GLP-1/Glucagon dual agonist — metabolic health and fat loss

DOSE	3–9 mg SC weekly
DURATION	24–48 weeks
PROTOCOL	Phase 3 trials ongoing; China-approved; weekly SC injection

Survodutide

GLP-1/Glucagon agonist — obesity and NASH treatment research

DOSE	0.6–6 mg SC weekly
DURATION	46 weeks (Phase 2)
PROTOCOL	Dose escalation protocol; under Phase 3 investigation

Orforglipron

Oral GLP-1 agonist — weight loss without injection

DOSE	12–45 mg oral daily
DURATION	36+ weeks (Phase 3 trials)
PROTOCOL	Non-peptide GLP-1 agonist; oral tablet; Eli Lilly Phase 3 ongoing

5-Amino-1MQ

NNMT inhibitor — fat loss, metabolism and muscle preservation

DOSE	50–150 mg oral daily
DURATION	4–8 week cycles
PROTOCOL	Oral capsule; often taken AM; cycle with breaks to maintain sensitivity

AOD-9604

HGH fragment — fat burning without blood sugar impact

DOSE	200–500 mcg SC daily
DURATION	4–12 weeks
PROTOCOL	Inject fasted AM or pre-workout; does not affect IGF-1

HGH Fragment 176-191

Isolated GH fat-loss fragment — targets adipose tissue

DOSE	250–500 mcg SC 2x/day
DURATION	8–12 weeks
PROTOCOL	AM fasted + pre-workout; does not raise IGF-1 or blood glucose

Tesamorelin

FDA-approved GHRH — reduces visceral fat, raises IGF-1

DOSE	1–2 mg SC daily
DURATION	26–52 weeks (FDA labeling for HIV-lipodystrophy)
PROTOCOL	Inject abdomen daily; FDA-approved; increases IGF-1 significantly

3. HEALING & RECOVERY PEPTIDES

BPC-157

Body protection compound — gut healing, tendon and tissue repair

DOSE	250–500 mcg SC daily
DURATION	4–12 weeks
PROTOCOL	Injection near injury site or systemic SC; oral for gut protocols

TB-500

Thymosin Beta-4 — accelerates healing and reduces inflammation

DOSE	5–10 mg SC weekly (loading); 2–5 mg maintenance
DURATION	4–6 weeks load, then monthly
PROTOCOL	Often stacked with BPC-157; loading phase 2–3x/week for 2 weeks

KPV

Anti-inflammatory tripeptide — gut healing and skin conditions

DOSE	200–500 mcg SC daily or oral
DURATION	4–8 weeks
PROTOCOL	Potent anti-inflammatory; IBD, psoriasis, wound healing protocols

GHK-Cu

Copper peptide — tissue regeneration, collagen and wound healing

DOSE	1–2 mg SC daily or topical
DURATION	4–12 weeks
PROTOCOL	SC for systemic effects; topical GHK-Cu creams for skin; anti-aging

GHK-Cu / KPV

Copper peptide + anti-inflammatory — healing and gut health

DOSE	0.5–1 mg SC daily
DURATION	4–8 weeks
PROTOCOL	KPV adds gut mucosal protection; often used for IBD/colitis protocols

AHK-Cu

Copper tripeptide — hair growth, scalp health and skin repair

DOSE	2–5% topical serum or 1 mg SC
DURATION	12–24 weeks for hair growth
PROTOCOL	Topical for scalp 2x/day; SC for systemic; often combined with GHK-Cu

LL-37

Antimicrobial peptide — immune defense and wound healing

DOSE	100–500 mcg SC or topical
DURATION	2–4 weeks
PROTOCOL	Cathelicidin; broad-spectrum antimicrobial; topical for wound protocols

Tri-Heal Blend

Triple healing stack — tissue repair, inflammation and recovery

DOSE	Per blend formulation
DURATION	4–8 weeks
PROTOCOL	Typically BPC-157 + TB-500 + KPV or similar; targeted injury protocol

Wolverine Stack

Comprehensive healing stack — rapid recovery and tissue repair

DOSE	Per blend formulation
DURATION	4–12 weeks
PROTOCOL	Multi-peptide healing protocol; often BPC-157 + TB-500 + extras

4. GROWTH HORMONE AXIS (GHRH / GHRP)

CJC-1295 DAC

Long-acting GHRH analog — sustained GH pulse elevation

DOSE	1–2 mg SC weekly
DURATION	8–12 weeks on, 4 weeks off
PROTOCOL	Once weekly; raises baseline GH over time; stack with Ipamorelin

CJC-1295 No DAC

Short-acting GHRH — mimics natural GH release pulse

DOSE	100–300 mcg SC 2–3x/day
DURATION	8–12 weeks on, 4 weeks off
PROTOCOL	Inject before sleep and/or pre-workout; pair with GHRP for synergy

Ipamorelin

Selective GH secretagogue — lean gains, minimal side effects

DOSE	100–300 mcg SC 2–3x/day
DURATION	8–12 weeks on, 4 weeks off
PROTOCOL	Cleanest GHRP; minimal cortisol/prolactin rise; best for first GH protocol

GHRP-2

GH releasing peptide — strong GH pulse, appetite stimulation

DOSE	100–300 mcg SC 2–3x/day
DURATION	4–8 weeks, cycle off 4 weeks
PROTOCOL	Potent hunger stimulant; inject fasted; stacks well with GHRH

GHRP-6

GH releasing peptide — muscle growth and hunger increase

DOSE	100–300 mcg SC 2–3x/day
DURATION	4–8 weeks
PROTOCOL	Strongest hunger stimulant of GHRPs; not ideal for fat loss phases

Hexarelin

Potent GHRP — strong GH release, cardiac and muscle protection

DOSE	100–200 mcg SC 2–3x/day
DURATION	4–8 weeks, then 4-week break
PROTOCOL	Strong GH release; desensitizes with prolonged use; cycle strictly

Sermorelin

GHRH analog — natural GH stimulation and anti-aging

DOSE	200–500 mcg SC before sleep
DURATION	6–12 months (anti-aging protocol)
PROTOCOL	Inject at bedtime on empty stomach; mimics natural GH pulse release

CJC/Ipamorelin 5/5

GH release blend (5 mg/5 mg) — fat loss and muscle support

DOSE	0.3 ml SC 2–3x/day
DURATION	8–12 weeks
PROTOCOL	Premix; inject before sleep + AM fasted; standard GH protocol blend

CJC/Ipamorelin 6/11

GH release blend (6 mg/11 mg) — enhanced GH output

DOSE	0.3 ml SC 2–3x/day
DURATION	8–12 weeks
PROTOCOL	Higher Ipamorelin ratio = stronger GH pulse; popular recomp blend

CJC/Ipamorelin 10/10

High-dose GH release blend — body recomposition

DOSE	0.3 ml SC 2–3x/day
DURATION	8–12 weeks on, 4 weeks off
PROTOCOL	High-dose version for experienced users; monitor for side effects

5. NOOTROPICS & COGNITIVE PEPTIDES

Semax

Nootropic — focus, memory, neuroprotection and BDNF boost

DOSE	100–900 mcg intranasal daily
DURATION	2–4 weeks on, 2 weeks off
PROTOCOL	Cyclical use; split AM/PM; stack with Selank for anxiety balance

Na Semax

N-acetyl Semax — enhanced bioavailability nootropic

DOSE	100–600 mcg intranasal daily
DURATION	2–4 week cycles
PROTOCOL	More potent than Semax; start low, assess cognitive response

Selank

Anxiolytic nootropic — reduces anxiety, improves mood and cognition

DOSE	250–750 mcg intranasal daily
DURATION	10–14 day cycles with breaks
PROTOCOL	Morning use preferred; may stack with Semax for cognitive synergy

Na Selank

N-acetyl Selank — longer-acting anxiety relief and cognitive support

DOSE	150–500 mcg intranasal daily
DURATION	2–3 week cycles
PROTOCOL	Longer half-life than Selank; once daily dosing often sufficient

PE-22-28

Antidepressant peptide — mood, cognition and neuroplasticity

DOSE	50–300 mcg intranasal daily
DURATION	4–8 weeks
PROTOCOL	BDNF-mimetic; intranasal preferred; research stage only

Cerebrolysin

Neuropeptide blend — cognitive function, stroke recovery and BDNF

DOSE	5–30 ml IV or IM daily
DURATION	10–20 day infusion courses
PROTOCOL	IV infusion protocol; used in neurological rehabilitation in 50+ countries

Cortagen

Brain cortex bioregulator — cognition and neurological health

DOSE	10 mg oral daily
DURATION	10-day courses 2x/year
PROTOCOL	Khavinson cortex peptide; stack with Semax/Selank for cognitive protocols

Pinealon

Pineal gland bioregulator — sleep, longevity and neuroprotection

DOSE	5–10 mg SC or oral daily
DURATION	10-day course, 2–3x/year
PROTOCOL	Khavinson peptide; melatonin pathway support; use before sleep

DSIP

Delta sleep-inducing peptide — sleep quality and stress reduction

DOSE	100–300 mcg SC or IV
DURATION	5–7 night courses as needed
PROTOCOL	Inject 30–60 min before sleep; reduces cortisol and ACTH

Kisspeptin

Hormone regulator — testosterone, LH and fertility support

DOSE	1–10 nmol/kg IV (research) or 50–100 mcg SC
DURATION	Acute protocol or pulsatile cycles
PROTOCOL	HPG axis stimulator; timing critical — often used as fertility aid

6. LONGEVITY & ANTI-AGING

Epitalon

Telomere lengthening — longevity and anti-aging

DOSE	5–10 mg SC daily
DURATION	10-day course, 2x/year
PROTOCOL	Khavinson protocol: 10 days on, 5.5 months off; may also use nasally

Na Epitalon

Sodium Epitalon — bioavailable epitalon variant for longevity

DOSE	5–10 mg SC or intranasal daily
DURATION	10-day course twice yearly
PROTOCOL	Same Khavinson protocol as standard Epitalon

NAD+

Coenzyme for energy, DNA repair and sirtuin longevity pathways

DOSE	250–500 mg IV or 500–1000 mg oral NR/NMN daily
DURATION	Ongoing or 4-week IV course
PROTOCOL	IV for acute repletion; oral NR/NMN for maintenance; with resveratrol

FOXO4-DRI

Senolytic peptide — clears senescent cells for anti-aging

DOSE	1–5 mg SC 3x/week
DURATION	3-day pulse protocol or 3-week course
PROTOCOL	Dasatinib-like senolytic; pulsed approach preferred over daily dosing

SS-31

Mitochondria-targeting antioxidant — energy and cellular protection

DOSE	10–40 mg SC daily
DURATION	4–8 weeks
PROTOCOL	Phase 2 trials in heart failure; research-grade only currently

MOTS-C

Mitochondrial peptide — metabolic health, insulin sensitivity and longevity

DOSE	5–10 mg SC 2–3x/week
DURATION	8–12 weeks
PROTOCOL	Exercise timing enhances effect; emerging longevity research peptide

SLU-PP-332

Exercise mimetic — activates muscle and metabolic pathways

DOSE	Research dosing; not standardized
DURATION	Research stage
PROTOCOL	ERR agonist; activates aerobic metabolic pathways without exercise

Abaloparatide

PTH analog — bone density and osteoporosis treatment

DOSE	80 mcg SC daily
DURATION	18–24 months max (FDA-approved limit)
PROTOCOL	FDA-approved for osteoporosis; injected in thigh/abdomen daily

Teriparatide

Parathyroid hormone analog — bone formation and fracture repair

DOSE	20 mcg SC daily
DURATION	24 months maximum (FDA labeling)
PROTOCOL	FDA-approved; self-inject daily; abdominal or thigh rotation

7. IMMUNE & ANTIMICROBIAL

Thymosin Alpha-1

Immune modulator — enhances immune function and antiviral defense

DOSE	1.6 mg SC 2x/week
DURATION	4–6 weeks (acute); longer for chronic conditions
PROTOCOL	Used in cancer support, Lyme disease, autoimmune; approved in 40+ countries

Thymulin

Thymus hormone — immune regulation and T-cell maturation

DOSE	50–100 mcg SC 2–3x/week
DURATION	4–8 weeks
PROTOCOL	Often cycled with Thymosin Alpha-1 for immune stack protocols

Vilon

Immune bioregulator — longevity and immune system modulation

DOSE	10 mg oral daily
DURATION	10-day courses twice yearly
PROTOCOL	Broad immune modulator; often part of longevity bioregulator stacks

Thymogen

Thymus bioregulator — immune system restoration

DOSE	100 mcg SC daily or 10 mg oral
DURATION	10-day course, 2x/year
PROTOCOL	Alpha-glutamyl-tryptophan dipeptide; immune aging reversal focus

Glutathione

Master antioxidant — detox, immune support and skin health

DOSE	600–1200 mg IV or 500 mg oral/sublingual daily
DURATION	4–8 weeks IV; ongoing oral
PROTOCOL	IV push 2–3x/week for skin; oral liposomal for maintenance

8. SEXUAL HEALTH & COSMETIC

PT-141

Bremelanotide — sexual arousal and libido enhancement

DOSE	0.5–2 mg SC/intranasal PRN
DURATION	As needed (not daily)
PROTOCOL	Inject 1–2 hrs before intimacy; FDA-approved (Vyleesi) for women

Melanotan I

Skin tanning and UV protection via melanogenesis

DOSE	0.5–1 mg SC daily
DURATION	2–4 weeks for tan maintenance
PROTOCOL	Sunlight exposure enhances effect; lower side effects than MT-II

Melanotan II

Tanning + libido enhancement + appetite suppression

DOSE	0.25–1 mg SC daily
DURATION	2–3 weeks loading, then 2–3x/week
PROTOCOL	Start low (0.1 mg) — nausea common; spontaneous erections possible

Oxytocin

Bonding hormone — social connection, stress relief and recovery

DOSE	10–40 IU intranasal PRN
DURATION	As needed; not for chronic daily use
PROTOCOL	Intranasal spray; social/recovery context; avoid dependency

SNAP-8

Wrinkle-reducing peptide — expression line relaxation (cosmetic)

DOSE	4% topical cream/serum
DURATION	4–8 weeks continuous use
PROTOCOL	Topical only; apply to forehead/crow's feet 2x/day

GLOW Blend

Skin and beauty blend — collagen, brightness and anti-aging

DOSE	Per blend formulation
DURATION	8–12 weeks
PROTOCOL	Often includes GHK-Cu + glutathione + collagen peptides; SC or topical

9. KHAVINSON BIOREGULATORS

Testagen

Testosterone bioregulator — supports natural T-level maintenance

DOSE	20 mg oral daily
DURATION	10-day courses, 1–2x/year
PROTOCOL	Khavinson bioregulator; usually 10-day course twice yearly

Prostamax

Peptide bioregulator — prostate health and male urinary function

DOSE	10 mg oral daily
DURATION	10-day courses 1–2x/year
PROTOCOL	Khavinson bioregulator; prostate gland extract-derived tetrapeptide

Ovagen

Liver bioregulator — hepatic function and detoxification support

DOSE	10 mg oral daily
DURATION	10-day courses 2x/year
PROTOCOL	Liver-derived tetrapeptide bioregulator; stack with NAD+ for hepatic support

Vesugen

Vascular bioregulator — blood vessel health and circulation

DOSE	10 mg oral daily
DURATION	10-day courses 2x/year
PROTOCOL	Vessel-derived bioregulator; cardiovascular anti-aging protocol

Bronchogen

Lung bioregulator — respiratory health and bronchial support

DOSE	10 mg oral daily
DURATION	10-day courses 1–2x/year
PROTOCOL	Khavinson lung peptide; useful post-illness respiratory support

Cartalax

Cartilage bioregulator — joint health and connective tissue repair

DOSE	10 mg oral daily
DURATION	10-day courses 2x/year
PROTOCOL	Cartilage-derived peptide; stack with BPC-157/TB-500 for joints

Chonluten

Bronchial and intestinal bioregulator — mucosal health

DOSE	10 mg oral daily
DURATION	10-day courses
PROTOCOL	Dual respiratory/GI bioregulator; useful for mucosal integrity

Livagen

Liver and immune bioregulator — DNA repair and immune support

DOSE	10 mg oral daily
DURATION	10-day courses 2x/year
PROTOCOL	Chromatin bioregulator — promotes gene expression for immune function

Cardiogen

Heart bioregulator — cardiac muscle health and protection

DOSE	10 mg oral daily
DURATION	10-day courses 2x/year
PROTOCOL	Cardiac tetrapeptide bioregulator; part of cardiovascular anti-aging protocol

Pancragon

Pancreatic bioregulator — blood sugar and digestive enzyme support

DOSE	10 mg oral daily
DURATION	10-day courses 1–2x/year
PROTOCOL	Pancreatic-derived; supports insulin regulation and enzyme production

Crystagen

Urinary system bioregulator — kidney and bladder health

DOSE	10 mg oral daily
DURATION	10-day courses 1–2x/year
PROTOCOL	Kidney/bladder-derived bioregulator; used for urinary tract health

11. SPECIALTY & BLENDS

VIP

Vasoactive intestinal peptide — inflammation, immune and gut health

DOSE	25–50 mcg SC/IV daily
DURATION	4–12 weeks depending on indication
PROTOCOL	Short half-life; often split doses AM/PM; used for MCAS/dysautonomia

L-Carnitine

Amino acid — fat metabolism, energy and cardiovascular support

DOSE	500–2000 mg oral daily or 1–2 g IV
DURATION	Ongoing supplementation
PROTOCOL	Oral with food or IV; L-ALCAR crosses BBB for brain benefits

PNC-27

Anticancer research peptide — selectively targets cancer cells

DOSE	Research protocols vary
DURATION	Research stage only
PROTOCOL	Penetratin-linked p53 peptide; highly experimental; preclinical only

Adamax

Neuroprotective stack — cognition, nerve repair and brain health

DOSE	Per blend formulation
DURATION	4–8 weeks
PROTOCOL	Vendor-specific blend; check individual peptide components and doses

IllumiNeuro Blend

Nootropic blend — focus, neuroprotection and mental clarity

DOSE	Per blend formulation
DURATION	4–8 weeks
PROTOCOL	Vendor-specific; typically intranasal or SC; assess tolerance first

KLOW Blend

Metabolic support blend — weight management and energy

DOSE	Per blend formulation
DURATION	8–12 weeks
PROTOCOL	Vendor-specific; typically includes GLP components or NNMT inhibitors

Tesa Blend

Tesamorelin blend — GH release and visceral fat reduction

DOSE	Per blend formulation
DURATION	12–26 weeks
PROTOCOL	Tesamorelin-based; daily SC injection; monitor IGF-1 levels

Neuroxelin Blend

Neural support blend — neuroprotection and cognitive enhancement

DOSE	Per blend formulation
DURATION	4–8 weeks
PROTOCOL	Vendor-specific neuro blend; check peptide components

10. Mitochondrial & Energy Protocols

SS-31 + NAD+ Mitochondrial Alliance

SS-31 (Elamipretide) directly protects the inner mitochondrial membrane via cardiolipin stabilization. NAD+ replenishes the cellular energy currency and drives SIRT1/SIRT3 sirtuin pathways. Used together they address both structural protection and metabolic fuel. Typical protocol: SS-31 20 mg SC daily + NMN 500 mg oral daily. 8-week course.

MOTS-c + NAD+ Metabolic Reboot

MOTS-c targets metabolic flexibility and gene expression via AMPK activation — acting like a signal that the body is exercising. NAD+ provides the substrate for sirtuin-based longevity pathways. Pair: MOTS-c 10 mg SC 3x/week + NMN/NR 500–1000 mg daily. Especially potent around exercise sessions.

SS-31 / MOTS-c / NAD+ 6-Week Progressive Protocol

Week 1–2: SS-31 10 mg SC daily + NMN 500 mg oral. Week 3–4: Add MOTS-c 5 mg SC 3x/week. Week 5–6: SS-31 20 mg, MOTS-c 10 mg, NMN 1000 mg. Take 4 weeks off. Repeat twice yearly for longevity focus.

12. Pinned Landmark Studies — BPC-157

The following are pinned landmark studies from the VPN Peptide Intel Feed — peer-reviewed, PubMed-indexed studies selected for their significance to BPC-157 research. Search each citation directly on PubMed to verify.

TENDONS & LIGAMENTS

BPC-157 Accelerates Full Tendon & Ligament Healing in Rat Models — Torn Achilles & Knee Ligaments Repaired

Staresinic M, et al. *J Orthop Res.* 2003 · PMID 12439832

Rats with surgically severed Achilles tendons and knee ligaments treated with BPC-157 demonstrated significantly faster and more complete healing compared to controls. Histological analysis showed superior collagen fiber organization, increased fibroblast activity, and restoration of tensile strength. BPC-157 promoted outgrowth of tendon fibers and reduced scar tissue — complete functional recovery was observed in treated animals within weeks of a tear that would otherwise cause permanent impairment.

OCULAR HEALING

BPC-157 Heals Corneal Lesions After Alkali Burn & Penetrating Eye Injury in Rats

Bilic I, et al. *Ophthalmologica.* 2005 · PMID 15802921

In rat models where the eye was subjected to alkali burns and penetrating puncture injuries, BPC-157 treatment dramatically accelerated corneal re-epithelialization and vascular healing. Rats in the BPC-157 group recovered near-complete corneal clarity and structural integrity. Controls suffered permanent scarring and visual impairment. BPC-157 promoted angiogenesis and healing in highly sensitive ocular tissue where standard treatments routinely fail.

KNEE REPAIR

BPC-157 Regenerates Bone-to-Tendon Junction — Medial Collateral & Cruciate Ligament Healing

Cerovecki T, et al. *J Orthop Res.* 2010 · PMID 20225288

BPC-157 administered to rats with completely transected medial collateral and anterior cruciate ligaments showed restoration of the bone-tendon insertion site — a notoriously difficult junction to heal — along with improved tensile strength, reduced fibrosis, and faster return of full range of motion. The mechanism involved upregulation of growth hormone receptor expression and enhanced VEGF-driven vascularization at the repair site. Controls showed persistent laxity and degenerative joint changes at follow-up.

TUMOR SAFETY — 2017

BPC-157 Injected Directly Into Mouse Tumors Did NOT Promote Tumor Growth — Safety Confirmed

Huang T, et al. *J Physiol Pharmacol.* 2017 · PMID 29151426

A critical safety concern for any angiogenic peptide is whether it accelerates tumor growth by promoting blood vessel formation in cancerous tissue. In this 2017 study, BPC-157 was injected directly into established tumors in mice. Results: tumor growth rates were not increased — and in several models, tumor markers trended lower in BPC-157 groups. The study concluded that despite BPC-157 pro-healing and angiogenic effects in normal tissue, it did not fuel malignant proliferation, offering an important safety signal for research applications.

13. Peptide Intel Feed — Research Highlights

These research highlights are sourced from the VPN Peptide Intel Feed — a curated, auto-refreshing feed of PubMed-indexed peer-reviewed studies across 27 peptide categories. Content is for educational reference only.

KPV

KPV Has a Self-Targeting Effect — Gut Healing Peptide That Finds Inflammation

KPV (Lysine-Proline-Valine) is an anti-inflammatory tripeptide derived from alpha-MSH. Inflamed gut tissue upregulates PEPT1 transporters, and KPV is preferentially absorbed through those same transporters — more inflammation means more KPV delivered exactly where it is needed. Research focus: IBD, IBS, Crohn's disease, ulcerative colitis, leaky gut, and psoriasis.

LARAZOTIDE ACETATE

Long COVID Research — Larazotide, Spike Protein & Gut Barrier

Ongoing clinical trials at Harvard/Mass General are investigating whether Larazotide Acetate can reduce spike protein leaking from the gut into the bloodstream, lowering systemic inflammation and Long COVID symptoms. BPC-157 + KPV + Larazotide = The Guardian Stack for gut healing and barrier protection. Also studied as adjunct in celiac disease to reduce abdominal pain, bloating, and diarrhea during accidental gluten exposure.

MOTS-C + SS-31

MOTS-c and SS-31 — Mitochondrial Sequence & Timing

MOTS-c targets metabolic flexibility and gene expression via AMPK activation while SS-31 (Elamipretide) directly protects the inner mitochondrial membrane via cardiolipin stabilization. Sequencing and timing matters. SS-31 is in Phase 2 trials for heart failure (EMPA-HEART). MOTS-c is emerging as a longevity research target with exercise-mimicking properties.

ATX304 (OS01)

ATX304 (OS01) + Tirzepatide — Theoretical Synergy for Weight Maintenance

ATX304 is an oral compound being studied as an AMPK and mitochondrial activator, researched alongside GLP-1/GIP agonists like Tirzepatide. Early research and anecdotal reports suggest people using ATX304 alongside or while titrating off a GLP-1 medication may be more likely to keep weight off afterward. PMC11981618 covers the ATX304 AMPK mechanism; PMC12545902 covers the GLP-1 combination research. Emerging MASLD/fatty liver data is also promising.

ELORALINTIDE + TIRZEPATIDE

Eloralintide + Tirzepatide = Bariatric Surgery Results — Phase 1b Data

Eli Lilly phase 1b trial data from EASD 2026: Eloralintide 9 mg + Tirzepatide 15 mg produced 29% weight loss. 17% in 16 weeks with just 3 mg Elora + 5 mg Tirz — comparable to what 2.4 mg semaglutide achieves in 72 weeks. The 29% arm matched Retatrutide's best results in a fraction of the time. Lilly's Enlighten-6 trial adds Eloralintide after GLP-1 plateau; Phase 2 in Type 2 diabetes wraps August 2026.

GLP-1 — REGULATORY UPDATE

6 Out of 7 Peptides Approved for Compounding — Landmark Regulatory Shift

6 out of 7 peptides reviewed have been approved for compounding by licensed 503A and 503B compounding pharmacies with a valid prescription. The 6 approved: BPC-157, TB-500, Ipamorelin, CJC-1295, PT-141. DSIP was the only compound denied — FDA cited insufficient clinical evidence in humans. Quality oversight for sterility, dosing accuracy, and purity are now regulated standards rather than guesswork.

GHK-CU

GHK-Cu — Systems-Level Skin & Tissue Regulator

GHK-Cu is a copper-binding tripeptide (Glycine-Histidine-Lysine) naturally found in human blood, saliva, and urine. It stimulates collagen, elastin, and glycosaminoglycan synthesis; activates antioxidant enzymes; promotes angiogenesis and nerve growth; and has been shown to regulate approximately 30% of human genes involved in tissue remodeling. In a lab dish test, 75-year-old human skin was functionally reverted to the chemical state of 30-year-old skin. Topical for skin; SC injection for systemic tissue repair. Dose: 1–2 mg SC daily or topical application.

THE GUARDIAN STACK

The Guardian Stack — Complete Gut Healing & Protection Protocol

BPC-157 (repair and regenerate) + KPV (calm IBD/IBS/colitis inflammation) + Larazotide Acetate (target zonulin pathway, tighten junctions, seal leaky gut) + Zinc Carnosine (protect the mucosa) + Quercetin or EGCG as Zinc ionophore (optimize zinc absorption and barrier integrity). Ideal for: leaky gut, IBS/IBD, inflammation, autoimmune support, and digestive disorders. Research references: NCBI PMC (Larazotide), PubMed, Stay Curious Metabolism (Substack).

14. Key Stacks & Protocols

The Guardian — Gut Healing & Barrier Protection

COMPOUNDS	BPC-157 + KPV + Larazotide Acetate + Zinc Carnosine + Quercetin/EGCG
DOSES	BPC-157: 250–500 mcg SC daily KPV: 200–500 mcg SC/oral daily Larazotide: per research protocol Zinc Carnosine: 75 mg 2x/day Quercetin: 500 mg daily
USE CASE	Leaky gut, IBS/IBD, Crohn's, colitis, autoimmune gut issues, Long COVID gut barrier repair.

The Wolverine Stack — Healing & Recovery

COMPOUNDS	BPC-157 + TB-500 (+/- KPV)
DOSES	BPC-157: 250–500 mcg SC daily TB-500: 5–10 mg SC weekly (loading), 2–5 mg maintenance KPV: 200–500 mcg SC optional
USE CASE	Accelerated tendon, ligament, muscle, and joint healing. Injury recovery, surgery prep/post-op.

The Kraken — Obesity & Metabolic Health

COMPOUNDS	Retatrutide + Eloralintide
DOSES	Retatrutide: start 2 mg SC weekly, titrate to 4–12 mg Eloralintide: phase trial dosing (3–9 mg per arm)
USE CASE	Maximum weight loss protocol. Four separate hormone pathways (GLP-1/GIP/Glucagon/Amylin) targeted simultaneously.

SS-31 + NAD+ Mitochondrial Alliance

COMPOUNDS	SS-31 (Elamipretide) + NMN or NR
DOSES	SS-31: 10–40 mg SC daily NMN: 500–1000 mg oral daily
USE CASE	Mitochondrial protection and energy production. Heart failure research, cellular aging, fatigue.

MOTS-c + NAD+ Metabolic Reboot

COMPOUNDS	MOTS-c + NMN/NR + resveratrol
DOSES	MOTS-c: 5–10 mg SC 2–3x/week NMN: 500–1000 mg daily Resveratrol: 500 mg daily
USE CASE	Metabolic flexibility, insulin sensitivity, longevity. Best paired with exercise protocol.

Semax + Selank — Nootropic Balance

COMPOUNDS	Semax + Selank (intranasal)
DOSES	Semax: 200–600 mcg intranasal AM Selank: 250–500 mcg intranasal AM
USE CASE	Focus + anxiety balance. Semax boosts BDNF and focus; Selank reduces anxiety without sedation.

CJC-1295 + Ipamorelin — Growth Hormone Support

COMPOUNDS	CJC-1295 (No DAC) + Ipamorelin
DOSES	CJC-1295 No DAC: 100–300 mcg SC Ipamorelin: 100–300 mcg SC — inject both before sleep
USE CASE	Natural GH pulse amplification. Fat loss, muscle preservation, sleep quality, anti-aging.

BPC-157 + KPV + TB-500 — Prostate Health

COMPOUNDS	BPC-157 + KPV + TB-500
DOSES	BPC-157: 250–500 mcg SC daily KPV: 200–500 mcg SC daily TB-500: 5 mg SC weekly
USE CASE	Research protocol for prostate inflammation, urinary symptoms, and tissue repair in the pelvic region.

Tirzepatide → Retatrutide Switch Protocol

COMPOUNDS	Tirzepatide then Retatrutide (never simultaneous)
DOSES	Last Tirz dose → wait 1–2 weeks → start Reta at 2 mg SC weekly → titrate up slowly
USE CASE	Safe transition between GLP compounds. Reta's added glucagon component means starting lower than your old Tirz dose.

Larazotide + BPC-157 — Gut Repair Combo

COMPOUNDS	BPC-157 + Larazotide Acetate
DOSES	BPC-157: 250–500 mcg SC daily or oral Larazotide: per trial dosing (12 mg/day oral in celiac trials)
USE CASE	Gut lining repair + tight junction sealing. Best combination for leaky gut and barrier dysfunction.

15. Blood Work Reference

Baseline blood work is non-negotiable before starting any research protocol. You cannot know if a compound is working, or if something needs attention, without knowing your starting numbers. Repeat every 1–3 months during active protocols.

Core Comprehensive Panel

- Complete Blood Count (CBC) with differential
- Comprehensive Metabolic Panel (CMP) — glucose, kidney, liver, electrolytes
- Lipid Panel — LDL, HDL, Total Cholesterol, Triglycerides
- HbA1c — 3-month blood sugar average
- Fasting Insulin — insulin resistance marker
- C-Reactive Protein (CRP) — systemic inflammation
- Ferritin — iron stores

Hormones

- Total & Free Testosterone
- Estradiol (E2)
- LH & FSH
- DHEA-S
- Prolactin
- SHBG (Sex Hormone Binding Globulin)
- IGF-1 — important baseline for GH peptide protocols
- Cortisol (AM fasted)

Thyroid

- TSH (Thyroid Stimulating Hormone)
- Free T3
- Free T4
- TPO antibodies (if autoimmune suspected)

Liver Panel

- ALT, AST, GGT

- Alkaline Phosphatase

- Total & Direct Bilirubin

- Albumin

Inflammation & Advanced Markers

- hs-CRP (high-sensitivity C-reactive protein)

- Homocysteine

- Uric Acid

- Vitamin D (25-OH)

- B12 & Folate

Editorial Design Audit — v1 vs v2

Category	Version 1	Version 2
Overall Design Score	v1: 38 / 100	v2: 84 / 100
Readability Score	v1: 31 / 100	v2: 88 / 100
Professional Appearance	v1: 42 / 100	v2: 86 / 100
Accessibility (WCAG)	v1: 29 / 100	v2: 91 / 100
Print Readability	v1: 15 / 100	v2: 90 / 100
Eye Fatigue (lower = worse)	v1: HIGH	v2: LOW
Home Printer Compatible	v1: NO	v2: YES

What Changed Between v1 and v2

Element	v1	v2
Background	Dark navy (#050012 body, #0e0625 cards)	White (#FFFFFF) throughout — fully printer-compatible
Body Text Color	#e8e0f0 on dark (contrast ratio ~4.2:1)	#2D2D2D on white (contrast ratio 10.8:1 — WCAG AAA)
Font Size — Body	9pt on dark backgrounds	9.5pt on white, justified, 14.5pt leading
Label Pills	Dark bar (#08041a) with colored text	Solid accent color pill — white text, WCAG AA compliant
Card Borders	1px colored glow borders on dark	0.8px #DDDDDD on white — clean, medical-journal style
Section Headers	Colored text on dark page	White text on solid color band — high contrast, clear hierarchy
Spacing Between Compounds	8px (too tight)	12px minimum — easier to scan
Body Text Alignment	Left-aligned	Justified — matches NEJM/Merck Manual standard
Eye Strain	High (light on dark, high contrast glare)	Low (dark on white, optimized leading)
Print Output	Solid dark backgrounds = wasted ink, unreadable	Pure white background — prints perfectly on home printer
Color Palette	Neon (#ff2d78, #00f5ff, #ffe600)	Medical deep tones (navy #1A4B8C, teal #006B6B, crimson #8B0000)

Accessibility

Fails WCAG AA for several text/bg combos

Passes WCAG AA and AAA for all body text